MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY, CHARGE STORAGE

TYPE 8931 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Direct-view charge storage tube, 5-inch flat face, self focusing, 2-writing guns, electrostatic focus and deflection, aluminized, P20 phosphor, high resolution and high contrast suitable for conventional radar presentation.

DIMENSIONS AND PIN CONNECTIONS: See figures 1 and 2

ABSOLUTE-MAXIMUM RATINGS: Writing guns - each gun (voltages with respect to writing gun cathode).

Parameter:	Ef	Ec1	ec1	Ec3	Ec2, Ec4	tk	ed	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	vdc (Note 3)	Vdc	Vdc	sec	vdc (Note 2)	Meg	Meg	Vdc	ft
Maximum:	6.93	0	2	1,500	1,500	---	400	1	0.1	+125	70,000
Minimum:	5.67	-200	---	---	---	30	---	---	---	-125	---
TEST CONDITIONS:	6.3	Adjust	---	10.8 ±.15% of Ec2, Ec4	900 to 1,100	---	Note 1	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS: Viewing gun (voltages with respect to viewing gun cathode).

Parameter:	Ef	tk	Ec5	Ec4	Ec6	Ec2, Ec3	Escn	Ehk	Rscn	Rbe
Unit:	V	sec	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Meg (Note 4)	Ohms (Note 5)
Maximum:	6.93	---	35	350	+150	175	11,000	+125	---	5 K
Minimum:	5.67	30	-20	---	-50	---	---	-125	1	---
TEST CONDITIONS:	6.3	---	+15	250	+75	70 to 125	10,000	---	1	---

GENERAL:

Qualification - Required (See note 42)
Aging - See notes 17(a) and 41

1/ Replaces tube type WX-5444

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1	48							
1301	Heater current (each gun)	-		---	---	If	0.280	0.320	A
5201	Electrode current (focusing) (each gun)	6	1,000 ftL	---	---	Ic3	-1	+1	μA
5201	Electrode current (cathode) (writing guns)	7		---	---	Ik	---	1.95	mA
5201	Electrode current (cathode) (viewing gun)	8		---	---	Ik	---	5.0	mA
5401	Voltage breakdown	-		---	---	---	---	---	---
5401	Voltage breakdown	9		---	---	---	---	No breakdown	---
---	Gas (viewing duration)	12		---	---	---	30	---	Sec
5101	Base alinement	Fig 1	3D4, base key	---	---	---	---	10.0	Degrees
5409	Blemishes	13, 14		---	---	---	---	---	---
---	Loose particles	15		---	---	---	---	---	---
5221	Light output	16		---	---	---	1,500	5,000	ftL
---	Writing speed (1) (both guns)	10	Ec1 = -10 Vdc (min)	---	---	---	36,000	---	in./sec
---	Writing speed (2) (both guns)	11		---	---	---	72,000	---	in./sec
---	Accelerating voltage (writing guns)	21		---	---	Ec2, Ec4	900	1100	Vdc
5406	Deflection factor (1D2 Gun B)	21		---	---	---	33.5	36.5	V/in.
5406	Deflection factor (1D2 Gun A)	-		---	---	---	33.5	36.5	V/in.
5406	Deflection factor (3D4 Gun A)	-		---	---	---	33.5	36.5	V/in.
---	Erasing	17		---	---	---	---	---	---
---	Resolution	18		---	---	---	65	---	Lines/in.
---	Electrode current (collimator)	14, 17(a)		---	---	Ic2, Ic3	---	1.0	mA
5402	Grid-cutoff voltage (writing guns)	19		---	---	Ec1	-75	-45	Vdc
---	Focusing voltage	19, 22	1,000 ftL	---	---	Ec3	10.65% of Ec2, Ec4	10.95% of Ec2, Ec4	Vdc
---	Suppressor cutoff voltage	20		---	---	Ec6	-10	0	Vdc
5407	Heater-cathode leakage	-	Ehk = ±125 Vdc	---	---	---	---	10	μAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1 - Continued								
5407	Electrode leakage (writing guns) (grid No. 1)	-		---	---	Ic1	---	1	μAdc
5407	Electrode leakage (writing guns) (grid No. 3)	-		---	---	Ic3	---	1	μAdc
---	Insulation of electrodes	24		---	---	---	100	---	Meg
---	Screen current	25		---	---	Iscn	---	700	μA
---	Heater-cathode warmup time	26		---	---	---	---	60	Sec
---	Grid response requirements (writing guns)	14, 27		---	---	---	20	---	%
---	Writing uniformity	14, 27		---	---	---	---	1.5	Ratio
---	Useful scan	-		---	---	---	3.5	---	Inches
---	Useful viewing diameter	14, 28		---	---	---	4.0	---	Inches
---	Vibration	45		---	---	---	---	---	---
---	Stable operation	47		---	---	---	---	---	---
	Quality conformance inspection, part 2	44, 46							
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---	---	---
5101	Angle between traces	29		1.0	II	---	89	91	Degrees
5231	Spot displacement (leakage)	23		1.0	II	---	---	1.0	mm
5404	Spot position (electrostatic deflection)	-		1.0	II	---	---	15	mm
5406	Deflection factor uniformity	-		1.0	II	---	---	5.0	%
---	Pattern distortion	30		1.0	II	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3	43							
---	Life-test provisions	14, 17(A), 44	Group C; light output= 1,800 ftL (min); t = 1,000 hours (min)	---	---	---	---	---	---
---	Life-test end point:								
	Light output	16		---	---	---	50	---	%
	Periodic-check tests	46							
---	Interaction factor	31		---	---	---	---	$1x10^{-5}$	in./Vdc
---	Magnetic shielding	32		---	---	---	---	0.5	mm
1331	Direct-interelectrode capacitance	-	Cathode to all, writing gun;	---	---	C	---	10	pF
			Grid No. 1 to all, writing gun	---	---	C	---	15	pF
			D1 to D2	---	---	C	---	3	pF
			D3 to D4	---	---	C	---	2	pF
			D1 to all	---	---	C	---	15	pF
			D2 to all	---	---	C	---	15	pF
			D3 to all	---	---	C	---	15	pF
			D4 to all	---	---	C	---	15	pF
			Suppressor grid to all	---	---	C	---	350	pF
---	Low-temperature operation	34		---	---	---	---	---	---
---	Low pressure-high temperature	36		---	---	---	---	---	---
---	Z axis cross talk	33		---	---	---	---	---	---
1006	Salt spray	40	No corrosion	---	---	---	---	---	---
1042	Shock	38, and Fig 4		---	---	---	---	---	---
---	Gray scales	17		---	---	---	7	---	---
---	Vibration	37		---	---	---	---	1.0	mm
---	Temperature cycling	39		---	---	---	5	---	Cycles
---	High-temperature operation	35		---	---	---	---	---	---
1002	Barometric pressure, reduced	-		---	---	---	---	---	---

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NOTES:

1. Guns A and B deflection plates shall have the mean deflecting electrode potential (100 ±3 vdc) applied simultaneously.
2. The peak voltage between accelerating electrode (g2, g4) and any deflecting electrode is ed.
3. The peak voltage between grid No. 1 (g1), and cathode is ec1.
4. Series current-limiting resistor used externally in screen circuit is Rscn. Voltage specified is measured at the supply end of resistor.
5. Total external circuit resistance in series with the backing electrode is Rbe.
6. Focusing electrode current. Screen brightness obtained by writing while erasing.
7. Cathode current (write gun).
   (a) With direct view storage tube (DVST) operated according to note 14 and voltages supplied to the deflection plates which will produce a large raster (to prevent damage to storage surface), the g1 of one writing gun shall be adjusted to 0 volts with the other writing gun biased beyond cutoff. Cathode current shall be within limits specified.
   (b) Repeat note 7(a), except that the writing gun previously measured shall be biased beyond cutoff, and the other writing gun then adjusted to 0 volts. Cathode current shall be within limits specified.
8. Cathode current (view gun). The DVST shall be operated according to note 14. The viewing gun storage surface shall be primed according to note 13(b)(2), (but not erased), to saturation brightness. The viewing gun cathode current under these conditions shall not exceed the value specified. Both writing guns are cutoff for this test.
9. Voltage breakdown. With the writing guns cutoff under test conditions, operate viewing gun under the following conditions for a minimum of 5 minutes:

Ef:	6.3 volts	Ec5:	-20 Vdc
Ec2, Ec3:	175 Vdc	Ec6:	+150 Vdc
Ec4:	350 Vdc	Escn:	11,000 Vdc
		ed:	400 Vdc

   At the completion of the specified time Escn shall be increased to 14,000 Vdc for 60 seconds. There shall be no recurrent breakdown between elements.
10. Writing speed (1).
    (a) Set up DVST according to test conditions. Adjust erase pulse for 3.5 ±0.5 seconds as described in note 17. No blanking pulse need be applied to g6. Apply a raster pattern to the tube by connecting a 60 Hz linear sawtooth voltage to the vertical axis and a 10 kHz linear sawtooth voltage to the horizontal axis of one of the writing guns. Retraces are to be blanked by a suitable writing gun intensity pulse train. Set the horizontal trace length to 3.50 inches (88.90 mm). Adjust the vertical scan to 2.75 inches (69.85 mm) and write to 1,200 ftL minimum. Measure light output anywhere on the face of the tube except dead center and record reading. Remove writing gun intensity pulse.

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NOTES: - Continued

10. Writing speed (1). - Continued

(b) Wait 30 seconds or more, then remove the erase pulse. With the writing gun intensity pulse adjusted to a judicious value more than 10 volts below zero bias, initiate the writing of one complete frame. Observe the reading of the light meter (which has not been moved from its initial position). If the reading is 90 percent minimum of the initial recorded reading, the test is complete. If not, reapply the erase pulse, wait 30 seconds or more. Adjust the writing gun intensity pulse to a higher value (still more than 10 volts below zero bias), remove the erase pulse and repeat the writing of one frame. Repeat, if necessary, until the light meter reads 90 percent of initial reading. If it is necessary to adjust the writing gun intensity pulse to less than 10 volts below zero bias to obtain the 90 percent minimum of initial reading, the tube has failed.

11. Writing speed (2).

(a) Repeat note 10(a) except vertical frequency shall be 120 Hz and horizontal frequency shall be 20 kHz.
(b) Repeat note 10(b) except that the acceptance level shall be 500 ftL minimum.

12. Gas (viewing duration). Write the viewing screen to saturation brightness; increase the erase pulse amplitude to just effect visual cutoff; remove the erase pulse; no area shall reach 10 percent of saturation brightness within 30 seconds as measured with a light meter having a 0.5-inch (12.70 mm) viewing diameter. Incident light on the area under test shall be less than 1 percent of saturated light output.

13. Blemishes. This test is performed with the DVST set up using the equipment and procedure described by note 27 and note 17. Adjust writing gun g1 of the gun under test to visual cutoff by adjusting the pedestal only and then adjust the gain/amplitude of the gated video noise input to obtain 20 percent of saturated light output as determined by use of the light meter in figure 3.

(a) Dark spots. Dark spots are defined as areas of the screen which do not brighten during writing or which require unusual processes to effect brightening; areas that lack persistence or whose persistence is obviously less than that of surrounding areas to an observer and includes the more easily recognized dead spots, burned spots and areas damaged by arc-overs, etc. The maximum number of dark spots is as follows, (refer to figure (blemish zones):

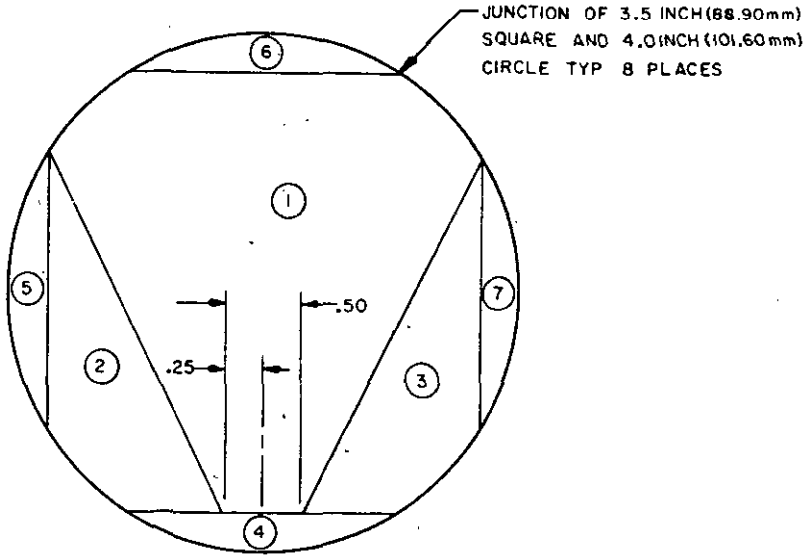
Maximum diameter of spot	Number of spots in zone	
	Zone 1	Zones 2 and 3
0.015 inch to 0.040 inch (0.38 mm) (1.02 mm)	2	4 total

The total number of dark spots shall not exceed 5 in zones 1, 2, and 3. Dark spots are permissible in zones 4, 5, 6, and 7 but shall not influence zones 1, 2, or 3. Dark spots less than 0.015 inch (0.38 mm) are permissible provided their separation is 0.3 inch (7.62 mm).

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NOTES: - Continued

13. Blemishes. - Continued



BLEMISH ZONES

(b) Light spots.

(1) Adjust the persistence of the tube to 3.5 ±0.5 seconds by adjusting the amplitude of the erase pulse. The erase pulse is to be a positive rectangular pulse, 66.0 ±10 percent microseconds wide at a pulse repetition frequency (PRF) of 225 pps. The erase pulse is coupled to the collimator according to note 28.
(2) Prime the tube. Priming is defined as momentarily switching the backing electrode from its normal circuitry to the collector and then back to its normal circuitry.
(3) Observe all bright spots 3.5 ±0.5 seconds after returning the backing electrode to its normal circuitry. Note their location and wait 30 seconds.
(4) Spots which do not erase in 30 seconds shall be defined as brilliant spots.
(5) Spots which erase in less than 30 seconds shall be defined as bright spots.

		Number of spots in zone	
Type	Maximum diameter of spot	Zone 1	Zones 2 and 3
Brilliant	Any Size	None	None
Bright	Greater than 0.030 (0.76 mm)	None	None
Bright	Equal to or less than 0.030 (0.76 mm)	2 0.5-inch (min) (12.70 mm) separation	4 total 0.25-inch (min) (6.35 mm) separation

The total number of light spots shall not exceed 5 in zones 1, 2, and 3. Bright spots in zones 2 and 3 shall not influence zone 1.

Light spots are permissible in zones 4, 5, 6, and 7 but shall not influence zones 1, 2, or 3.

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NOTES: - Continued

14. Test conditions. Unless otherwise specified, test conditions shall be as specified on page 1.

15. Loose particles. While no clearly defined rules can be offered for loose particle content in a DVST, the following shall apply:

    (a) Tubes which exhibit bursts of momentary particle activity on the application of any voltages, and in which particles show no clinging action so as to cause blemishes, shall be considered acceptable.
    (b) Tubes which exhibit particle activity on the application of voltages which cause blemishes, arcing, repeated flashing, and other disturbances shall not be acceptable.

16. Light output. Measured at saturation brightness. Light output is measured over any 1/2 inch (12.70 mm) or greater area, any place within the useful viewing diameter. Saturation is the level of brightness obtained several seconds after cessation of writing with no erase pulse applied. Life-test end point shall be 50 percent of initial saturated brightness of the life-test sample.

17. Erasing.

    (a) Erasure shall be accomplished by applying positive rectangular pulses to the backing electrode. Incident light on the area under test shall be less than 1 percent of saturation brightness. Write the useful screen area to saturation brightness. Apply erase pulses of width 66.0 ±10 percent μs, and PRF of 225 pps. Erase time of 3.5 ±0.5 seconds shall be obtainable by adjusting the erase pulse amplitude to a single value within the range 2 to 12 volts peak-to-peak.
    (b) This test is performed with an 85 volt negative pulse applied to the suppressor grid (g6). The pulse shall be of approximately the same pulse width as the erase pulse and in sync with the erase pulse. The rise and the fall time of the suppressor pulse shall be adequate to suppress the unwritten background light level to be less than 5 ftL.

18. Resolution. To measure resolution in terms of lines/inch at the same time that the light output is equal to 1,000 ftL, the procedure is as follows:

    (a) Adjust erase time according to note 17. Apply a 45 to 80 line raster pattern to the tube by connecting a 60 Hz sawtooth voltage to the vertical and 2,700 to 4,800 Hz sawtooth voltage to the horizontal deflection plates of one of the writing guns. (Note: The selection of a specific number of lines will be determined by the size of the area measured and by the type of light meter being used). Set the raster trace length at 3-1/2 inches (88.90 mm).
    (b) Expand the low-frequency scanning amplitude. Compress the raster until the lines just merge (line structure disappears). A photometer, corrected for the spectral response of the eye and having the diameter of the aperture of its light sensitive area smaller than the most narrow dimension of the collapsed raster, shall be used to measure the light output. Set the light level at 1,000 ±5 percent ftL at the center of the raster.

        NOTE: It may be necessary to further collapse or expand the raster by changing the amplitude of low-frequency scan while adjusting light output in order to obtain a properly merged raster at the desired brightness level).

    (c) Again expand the pattern. Compress the raster until the lines just merge at the center of the pattern. At the proper degree of merging the distinguishable line structure just disappears. If the raster is overmerged, there will result a reverse line structure consisting of bright lines where the dark lines were prior to merging. Measure the width of the compressed pattern at the center of the raster. The resolution is then given by the quotient of the number of lines in the raster divided by the measured width of the compressed pattern in inches. Repeat the above procedure at points on the vertical axis 1-1/2 inches (38.10 mm) above and 1-1/2 inches (38.10 mm) below the center of the tube face. Interchange the connection of deflection plates. Re-measure resolution, at the center of the tube and at points on the horizontal axis 1-1/2 inches (38.10 mm) right and 1-1/2 inches (38.10 mm) left of the center of the tube face. All of the resolution values calculated from the measurements described above shall be greater than or equal to the minimum value specified.

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NOTES: - Continued

18. Resolution. - Continued

    (d) Repeat the above procedure for the other writing gun.

19. Grid cutoff voltage (writing gun). All voltages shall be with reference to the cathode of the gun under test. Other voltages are according to test conditions in note 14. Erase pulse set according to note 17.

20. Suppressor cutoff. With the backing electrode at, or greater than, the positive peak value of an erase pulse required to cut the viewing gun off, (all other voltages at test conditions) the maximum light output of any area of the screen shall be less than 1.0 ftL.

21. Acceleration voltage.

    (a) Adjust accelerating voltage on gun B for a deflection factor of 35V/inch on the 3D4 plates of gun B. This voltage will be used on gun A and B for all other tests in this specification.
    (b) Use the same accelerating voltage that was determined in note 21(a) for both gun B (1D2) and gun A (1D2, 3D4) deflection factors.

22. Focusing voltage. Set accelerating voltage according to note 21. The tube shall meet all performance requirements with the focus voltage, (Ec3 with respect to writing gun cathode), at any value between 10.65 percent and 10.95 percent of accelerating voltage (Ec2, Ec4).

23. Spot displacement. Adjust erase pulse to a convenient value. Connect each deflecting electrode through a separate 100 K ohm resistor to g2, g4 of the writing gun. With means for short circuiting each resistor, measure the spot displacement when each resistor is short circuited in turn.

24. Insulation of electrodes.

    (a) The dc resistance, between any element including the magnetic shield and all other elements connected together shall be no less than specified. The backing electrode (g5) and suppressor (g6) shall be measured at a potential of 150 Vdc. All other elements shall be measured at a potential of +250 Vdc. The heaters shall not be subjected to this test.
    (b) The dc resistance, between the flood gun cathode or both writing gun cathodes, and all other elements connected together shall be no less than 15 megohms.
    (c) Should the polarity of the measuring potential influence the resistive reading, reverse the polarity and the lower of the two readings shall apply. The heater shall not be energized during this test.

25. Screen current. Test performed with entire viewing area illuminated to maximum saturation brightness.

26. Heater-cathode warmup time. After 60 seconds of heater-cathode warmup period, application of normal tube voltage shall produce a collimated 4-inch (min) (101.60 mm) display (with writing gun on).

27. Grid response. During the grid response requirements tests, performance is judged by feeding a random noise signal of known amplitude to each of the two writing guns in succession and measuring the resulting display luminescence as a percentage of saturated light output. The writing uniformity test shall be performed immediately following and in conjunction with grid response. The equipment to be used, the conditions of test, the calibration procedure, and the sequence of test operations are described as follows:

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NOTES: - Continued

27. Grid response. - Continued

(a) Equipment.

(1) Random noise generator (5 MHz scale) - General Radio Type 1390-B, or equivalent.
(2) Sine wave generator (2, 000 to 10, 000 Hz) - Hewlett-Packard Model 200 ABR, or equivalent. (Note: Sine wave shall be non-synchronous with vertical pulse repetition frequency (PRF)).
(3) Video gating and mixing circuitry designed to accept, successively, the output from the random noise and sine wave generators. These signals shall be gated on only during the mutual period of the vertical scan and pedestal. Negative-going portions of noise shall be rejected; i. e., clipped. Noise and pedestal shall add. Do not clip calibrating sine wave.
(4) Appropriate scanning waveform, pedestal, and erase pulse generators. (See the following conditions of test).
(5) Noise calibration circuits according to figure 3.
(6) Light output measuring device according to figure 3.

(b) Conditions of test.

(1) Vertical scan height: 3.5 inches (88.90 mm) (centered).
(2) Vertical scan time: 625 μs. (linear sawtooth).
(3) Vertical PRF: 500 pps.
(4) Horizontal scan width: 3.5 inches (88.90 mm) (centered).
(5) Horizontal scan rate: 3.5 in./sec (88.90 mm) (linear triangular).
(6) Mean deflecting electrode potential: 100 ±3 Vdc.
(7) Writing gun g1 input: Output waveform from noise calibrating circuitry, clamped to g1 bias level.
(8) Noise calibrating video input circuitry to point A.

a. Pedestal only (625 μs rectangular pulse coincident with vertical scan period).
b. Pedestal plus gated sine wave, or,
c. Pedestal plus gated and clipped noise.

(9) Noise calibrating circuitry plus input to point B; pedestal only.
(10) Erase pulse; positive going rectangular pulse, 66.0 μs ±10 percent wide at PRF of 225 pps.
(11) Brightness pulses applied to g6 according to note 17(b).
(12) Other voltages according to test conditions.

(c) Calibration procedure.

(1) With pedestal only applied to point A, adjust pedestal amplitude for 10 volt peak-to-peak reading at point Eo of figure 3. No further adjustment permissible.
(2) With 12 volt peak-to-peak pedestal at input B, adjust R7 of noise calibrating circuitry for null reading on meter M1.
(3) Apply sine wave to pedestal, i. e., pedestal plus gated sine wave should be applied to point A. (Note: Amplitude of pedestal itself shall be same as in step (c)(1). Adjust the amplitude of the gated sine wave to be equal to 1/2 (measured grid cut-off voltage) -5 volts peak-to-peak as indicated by a calibrated scope at point Eo of figure 3. (The maximum allowable voltage is to be 23 volts peak-to-peak.) Read and record meter M1. Remove gated sine wave from pedestal.
(4) Cover opening to photoelectric cell V3, close switch S1, and adjust potentiometer R15 for null reading of meter M2.

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NOTES: - Continued

27. Grid response. - Continued

(c) Calibration procedure. - Continued

(5) Write display to saturated brightness by reducing erase pulse amplitude to zero, adjusting writing bias so that pedestal writes fully a "stored" blank raster, and adjusting write bias to cutoff.
(6) Place photoelectric cell V3 against the faceplate of DVST and after a steady reading is obtained, read and record meter M2.

NOTE: Avoid saturation of the measuring device by the addition of sufficient neutral density filters in front of the photoelectric cell. A reading of approximately 4.5 volts indicates saturation. Sufficient filtering to obtain a reading between 3 and 4 volts will yield a satisfactory measurement.

(d) Test sequence.

(1) Apply erase pulse as in (b)(10). Amplitude shall be between 2 to 12 volts peak-to-peak.
(2) Adjust bias of writing gun g1 under test, with pedestal only, until meter, M2 reads the same as step (c)(6).
(3) Turn off writing gun when sweep is in center and note the time required for saturated light output to be erased to background brightness (complete erasure over entire face of tube).
(4) Repeat steps (d)(2) and (d)(3) adjusting erase pulse amplitude to obtain 3.5 ±0.5 seconds erase time.
(5) Apply pedestal only to point A. Adjust bias of writing g1 under test just to visual cutoff.
(6) Apply pedestal plus gated and clipped noise to point A. (Note: Amplitude of pedestal itself shall be same as in step (c)(1). Adjust amplitude of noise until meter M1 reads the same as in (c)(3).
(7) Place photoelectric cell against faceplate of DVST. Close S1 momentarily. Release switch S1 and allow sufficient time for integration of noise to reach equilibrium. Scan the face of DVST with photocell to find and record the minimum and maximum readings of M2.
(8) The minimum reading is step (d)(7) shall be no less than 20 percent of the meter reading in step (c)(6).
(9) If the minimum reading observed in step (d)(7) is less than 50 percent of the meter reading in step (c)(6), the ratio of the maximum to minimum reading in step (d)(7) shall not exceed the value specified for writing uniformity.
(10) If the minimum reading in step (d)(7) is greater than 50 percent of the meter reading in step (c)(6), reduce the calibrated noise input to obtain a value which is 40 percent of the meter reading in step (c)(6). Again scan the face of the DVST for minimum and maximum values. The ratio of maximum to minimum readings shall not exceed value specified for writing uniformity.
(11) At the completion of step (d)(10) remove noise input and close S1 momentarily. Verify that the erase time is still 3.5 ±0.5 seconds as originally setup in step (d)(4).

28. Useful viewing diameter. The external circuitry required to prevent buildup of a bright ring around the periphery of the collimated viewing area shall consist of no more than coupling the erase pulse through a suitable capacitor to the collimating electrode. The erase pulse shall be the same as described in note 17. The luminance of the bright ring shall not exceed 20 ftL.

29. Angle between traces. When measured with a device having an accuracy of at least ±0.2 degrees, the angle between horizontal and vertical traces shall be within the limits specified.

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NOTES: - Continued

30. Pattern distortion. The total movement of the ends of a 2.8 inch (71.12 mm) trace (centered with respect to the tube face), deflected horizontally or vertically 1.40 inches (35.56 mm) each side of the center of the tube face, shall not exceed 0.060 inch (1.52 mm).

Alternate method for measuring pattern distortion. A rectangular raster shall not fall outside a square frame whose inside dimension is 2.74 inches (69.60 mm) and outside dimension is 2.86 inches (72.64 mm) and whose center coincides with the centerline of the tube.

31. Interaction factor. The deflection of one beam, with balanced +250 Vdc voltages applied to deflection electrodes of the other beam, shall not be greater than the value specified.

32. Magnetic shielding. The shielding shall be such that a 1.3 gauss field will not deflect an undeflected spot more than 0.5 mm and will not otherwise affect performance of the tube described in this specification. The magnetic shield shall be secured to the tube at least to the full length of zones A, B, and C, as shown by figure 1, with an encapsulating material, electrically non-conducting and non-nutrient. The shield shall be capable of withstanding a magnetic field when tested by the following or equivalent procedure. A suitable pair of air-core coils to produce the magnetic field for testing has 50 ampere-turns each, a mean crossection of 10.5 x 2.75 inches (266.70 x 71.12 mm) perpendicular to the field and a separation of 8.5 inches (215.90 mm). The center of the coils shall be approximately centrally located about the tube. With the tube operating as for the spot position test, measure spot motion when the coil current is interrupted. The spot motion under these conditions shall not exceed 0.5 mm.

33. Capacitance. The capacitance of g1 to any deflection plate shall not exceed 5.6 pF. The difference between the capacity of any deflection plate of a pair and the other plate of the same pair shall not exceed 1.4 pF. All elements except those under test may be grounded.

34. Low-temperature operation. Tube shall be cold soaked for a period of 48 hours at -65°C. At the end of this period and while still at the low temperature, all heater voltages shall be applied for a period of 15 minutes. Tube shall then be removed from freezer and tested for gas (viewing duration).

35. High-temperature operation. Place tube in chamber having an ambient air temperature of +150°C. Apply heater voltages and retain tube in chamber for a period of 6 hours. At the end of this period, remove the tube and test for gas (viewing duration). Tube shall be within initial limits.

36. Low-pressure-high-temperature operation. The tube shall be conditioned for a period of not less than 1 hour in a chamber having an ambient air temperature of 100°C. At the end of 1 hour, the tube shall be operated with normal voltages applied and at a pressure equivalent of 32,000 feet. Following this, the chamber temperature shall be reduced to +53°C. At the end of 1 hour the tube shall be operated with normal voltages applied and a pressure equivalent of 70,000 feet altitude. Connections shall be made to the base pins in a manner that does not degrade the tube breakdown characteristics. The criteria of satisfactory operation for a period of 15 minutes shall be the presence of a clearly defined circular trace display (absence of corona and arc-over) and g1 leakage shall not have changed by more than 2 μA from the initial measurement.

37. Vibration. The tube shall be suitable mounted on the vibration table in zones A and C (see figure 4) and tested as follows:

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NOTES: - Continued

37. Vibration. - Continued

(a) With the tube mounted and with a circular trace positioned on the tube face and operating at prevailing room temperature, the tube shall be vibrated along the X-X axis with the frequency varying between 10 and 500 Hz at an amplitude as indicated by vibration specification curve (see figure 4) as monitored at a point on the metal shield immediately adjacent to one of the mounting areas. The frequency change, as a function of time, shall be logarithmic, and shall be such that a complete cycle, for example, 10 to 500 to 10 Hz, will consume approximately 15 minutes. The test shall continue for 90 minutes. During this time, all resonant points shall be noted and the increase in linewidth of the circular trace shall be within the limits specified. To accurately determine resonant points, the frequency shall be varied slowly through the range around each of the noted resonant points. A resonant point is defined as that frequency at which a noticeable change in the written or stored information on the screen is visible. Tubes shall be rejected for occurrence of permanent brilliant or dark spots exceeding blemish limits, erratic (broken) trace, or for other indications of tube malfunction or degradation in performance.

(b) With the tube operating, it shall be vibrated for 30 minutes at the resonant frequency noted above. Where more than one resonant frequency is noted, the test period may be divided among the resonant frequencies, whichever is considered most likely to produce failure. Where resonant frequencies are not apparent within the specified frequency range, the tube shall be vibrated for a period of 1 hour at a frequency of 55 Hz. During this time the increase in linewidth of the circular trace shall be within the limits specified. Tubes shall be rejected for occurrence of permanent brilliant or dark spots exceeding blemish limits, erratic (broken) trace, or for indications of malfunction or degradation in performance.

(c) Repeat note 37(a and b) with the tube mounted such that the vibration is along the Y-Y axis.

(d) Repeat note 37(a and b) with the tube mounted such that the vibration is along the Z-Z axis.

(e) Following vibration, retest tube. All requirements and test limits specified in quality conformance inspections, part 1 (except vibration) and part 2 of this specification shall be satisfied. In the event a reject DVST has been used for this test, under the conditions outlined in note 44, the item which was cause for initial rejection may be disregarded.

NOTE: This test is classified as a destructive test, in the sense that tubes subjected to it shall not be shipped.

38. Shock. Tube shall be subjected to 15G, 11 millisecond duration shock; 3 shocks in each of the 3 mutually perpendicular axes (18 total); tube operating, verification of proper operation during each impact not required. Following the shock test, tube shall be tested for gas (viewing duration) and spot displacement.

39. Temperature cycling. All heaters shall be energized with nominal heater voltage during that portion of each cycle beginning at low-temperature equilibrium and ending at high-temperature equilibrium. Use MIL-STD-202, method 107, condition B, except temperature in step 3 shall be 100° ±3°C. Following temperature cycling, retest tube for gas (viewing duration).

40. This test is performed only during qualification inspection.

41. Aging. Prior to the performance of any tests, the tube shall be operated according to note 14 by the manufacturer, for a minimum of 48 hours.

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NOTES: - Continued

42. Qualification. All tubes submitted for qualification shall be subjected to quality conformance inspection, parts 1, 2, 3, rough handling test, and salt spray (corrosion).

    (a) Tubes submitted for qualification will not be shipped to the customer.
    (b) Rough handling test shall be in accordance with method 1136.
    (c) Salt spray (corrosion) shall be in accordance with method 1006.

43. From each group of 50 tubes produced (or from each monthly production lot, if the lot exceeds 50 tubes) select 1 tube to be evaluated against these tests. One tube may be subjected to all tests or a separate tube from the same production lot may be selected for each test whichever is more convenient. If this initial sample passes each required test, the group of 50 tubes (or the monthly lot) may be shipped. If a test is failed, two additional samples shall be subjected to that test and pass the test to permit release of the group which they represent. If any of these samples fail, 100-percent production evaluation is required until such time as 10-successive tubes pass.

44. Operable reject tubes from current lot may be used for these tests provided the outcome of the tests shall not be affected by the defect causing the rejection. The original defect shall be disregarded when determining whether the tube under test passes or fails these tests.

    NOTE: Tubes for life test may be rejects for mechanical or blemish defects only.

45. Vibration. This is a 100-percent test and shall be applied to all tubes. With the tube mounted as specified in note 37 and operating at prevailing room temperature, the tube shall be vibrated along the Y-Y axis with the frequency varying between 10 and 500 Hz at an amplitude as indicated by the vibration specification curve (same as in note 37). The frequency change, as a function of time, shall be logarithmic, and shall be such that a complete cycle (10 to 500 to 10 Hz) will consume approximately 15 minutes. The tube shall be vibrated for one cycle. During this period, maximum particle activity shall be observed and recorded in the lower and upper frequency ranges (10 to 100 Hz and 100 to 500 Hz). Select a single fixed frequency in each of the two ranges where particle activity was observed and vibrate the DVST for an additional 5 minutes at each of the frequencies. Tubes which display permanent blemishes on the screen, arcs or flashing, or develop permanent or temporary shorts, shall be rejected. Following vibration, retest the tube. All initial requirements and test limits shall apply and the tube shall be within initial limits.

46. Shipment of tubes prior to completion of quality conformance inspection, part 2 and part 3 (periodic-check tests) is permitted provided quality conformance, part 2 and part 3 (periodic-check tests) have been successfully completed on the preceding month's lot.

47. Stable operation.

    (a) At the beginning of the DVST final acceptance test sequence, measure and record the control grid cutoff voltage of each writing gun between 30 and 60 seconds after energizing writing gun heaters.
    (b) After a 20-minute period, during which other DVST final acceptance tests are performed, remeasure and record the control grid cutoff voltage of each writing gun. The difference between the cutoff value at 60 seconds and 20 minutes for each writing gun is the delta cutoff and shall be equal to or less than 3.0 volts.

48. Quality conformance inspection, part 1, tests shall be a 100-percent production evaluation test requirement.

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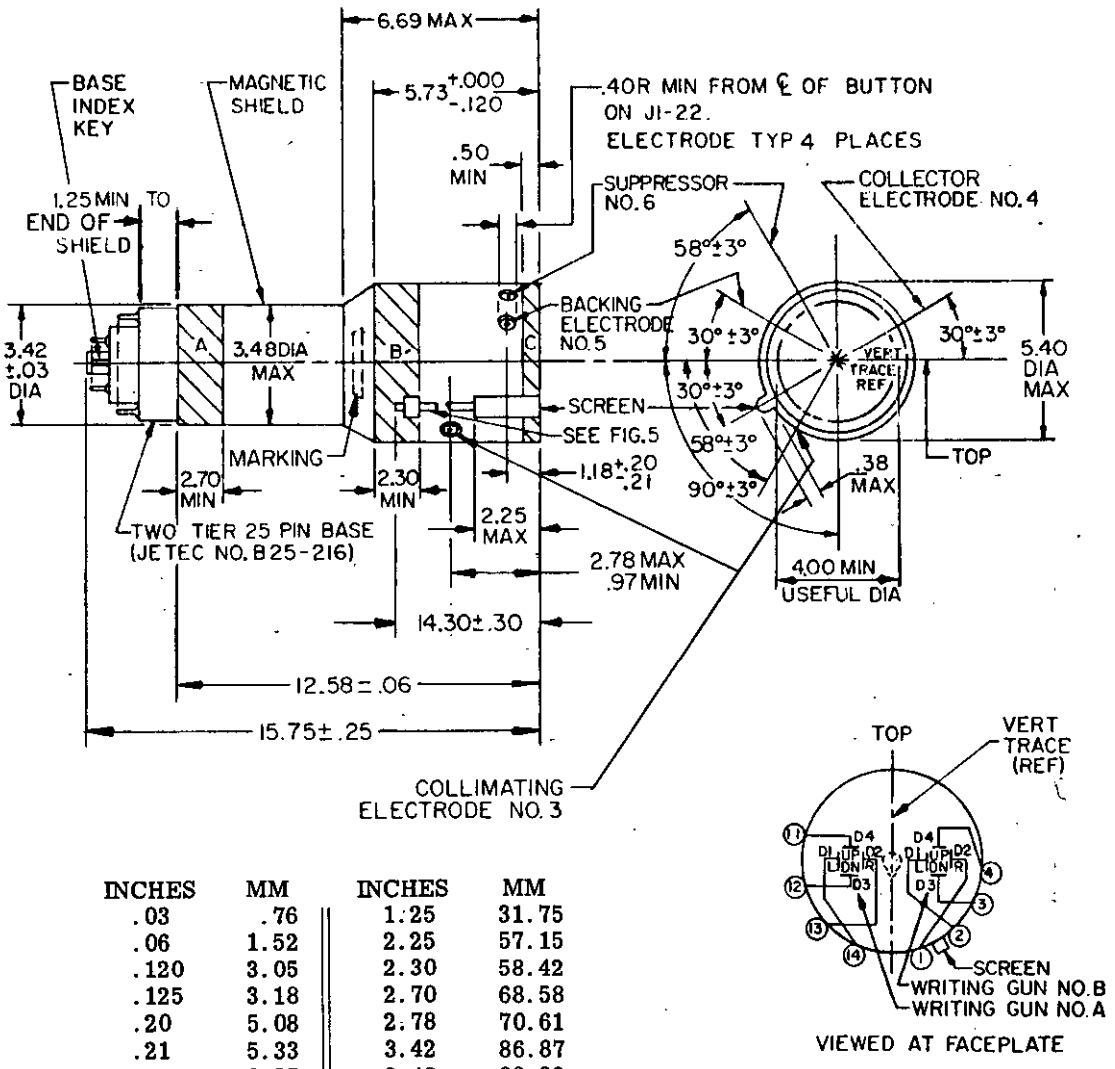
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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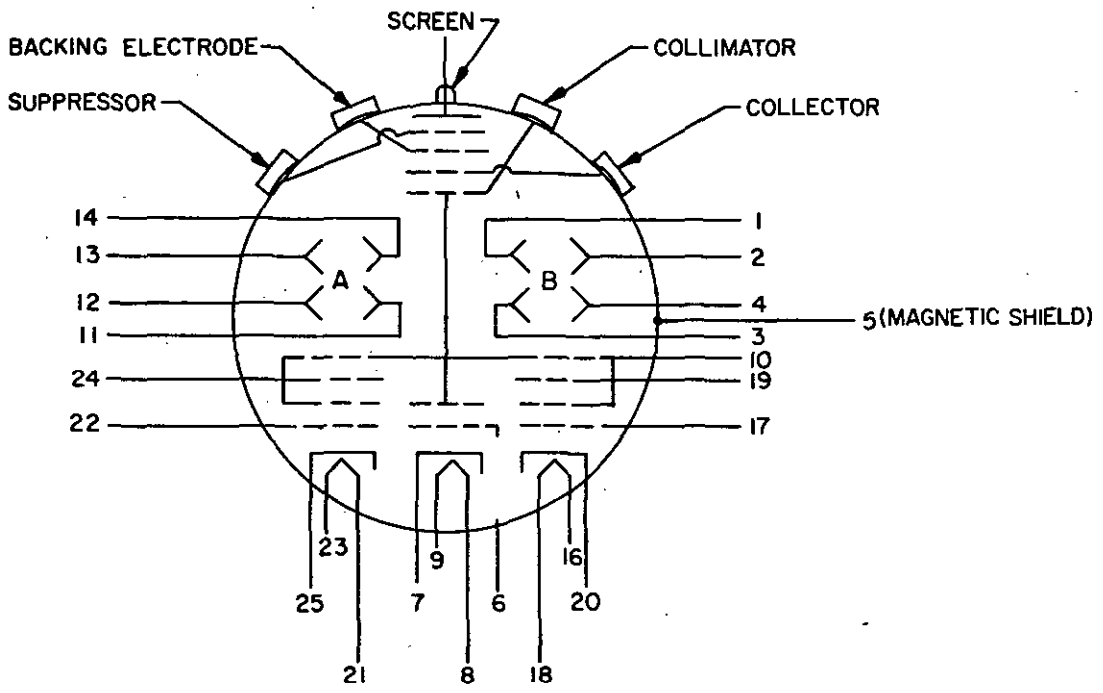


INCHES	MM	INCHES	MM
.03	.76	1.25	31.75
.06	1.52	2.25	57.15
.120	3.05	2.30	58.42
.125	3.18	2.70	68.58
.20	5.08	2.78	70.61
.21	5.33	3.42	86.87
.25	6.35	3.48	88.39
.30	7.62	5.40	137.16
.38	9.65	5.73	145.54
.40	10.16	6.69	169.93
.50	12.70	12.58	319.53
.97	24.64	14.30	363.22
1.18	29.97	15.75	400.05

NOTES:
a. Dimensions are in inches.
b. Unless otherwise specified, tolerances are .005 (.13 mm) for three place decimals and .02 (.51 mm) for two place decimals.
c. Weight of the tube is 5.25 ± 0.25 pounds.

FIGURE 1. Outline drawing of electron tube type 8931.

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Pin No. 1 Deflecting electrode D2 of writing gun B.
Pin No. 2 Deflecting electrode D1 of writing gun B.
Pin No. 3 Deflecting electrode D3 of writing gun B.
Pin No. 4 Deflecting electrode D4 of writing gun B.
Pin No. 5 Magnetic shield.
Pin No. 6 No connection. (See note b.)
Pin No. 7 Cathode of the viewing gun. (See note a.)
Pin No. 8 Heater of the viewing gun.
Pin No. 9 Heater of the viewing gun.
Pin No. 10 Accelerating electrodes (No. 2 and No. 4 of writing guns).
Pin No. 11 Deflecting electrode D4 of writing gun A.
Pin No. 12 Deflecting electrode D3 of writing gun A.
Pin No. 13 Deflecting electrode D2 of writing gun A.
Pin No. 14 Deflecting electrode D1 of writing gun A.
Pin No. 15 No connection.
Pin No. 16 Heater of writing gun B.
Pin No. 17 Control grid (No. 1) of writing gun B.
Pin No. 18 Heater of writing gun B.
Pin No. 19 Focusing electrode (No. 3) of writing gun B.
Pin No. 20 Cathode of writing gun B. (See note a.)
Pin No. 21 Heater of writing gun A.
Pin No. 22 Control grid (No. 1) of writing gun A.
Pin No. 23 Heater of writing gun A.
Pin No. 24 Focusing electrode (No. 3) of writing gun A.
Pin No. 25 Cathode of writing gun A. (See note a.)

NOTES:
a. The viewing gun control grid may be connected to pin 7 and pin 20 (or pin 25) through an internal voltage divider network.
b. Pin 6 may be connected to the viewing gun control grid. However, pin 6 is grounded externally to the tube.

FIGURE 2. Pin connections.

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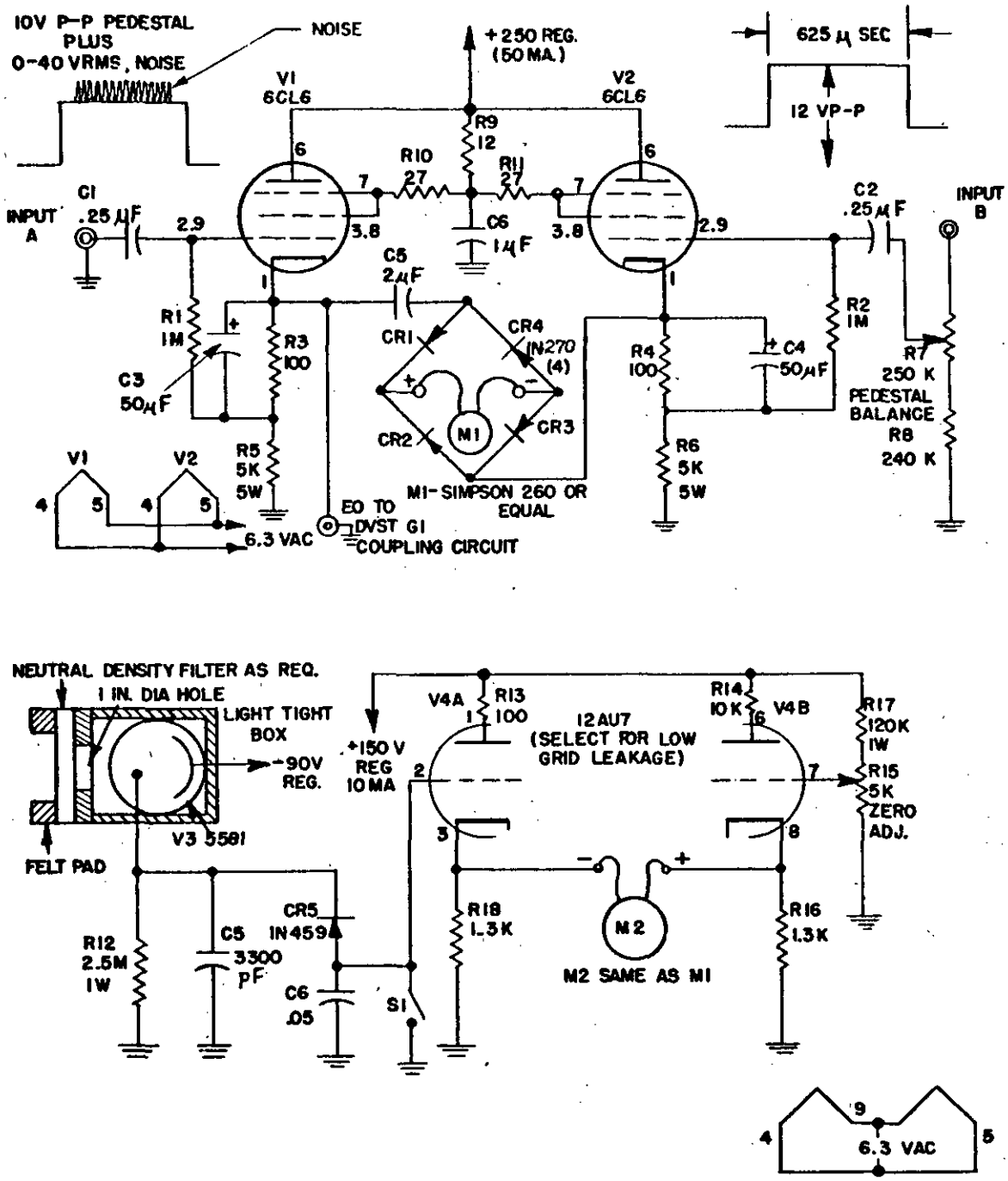
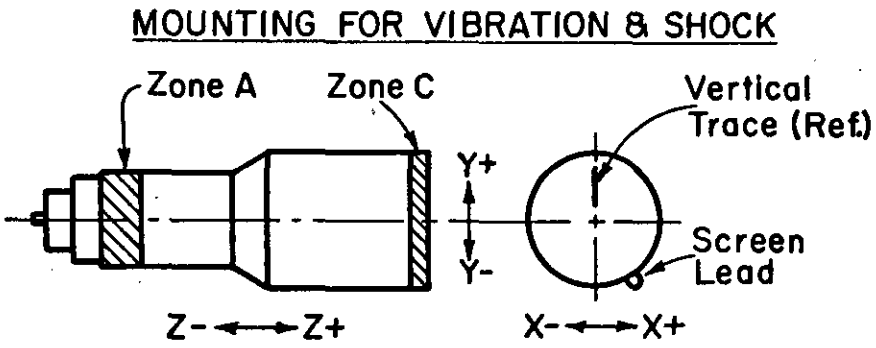


FIGURE 3. Grid drive test equipment.

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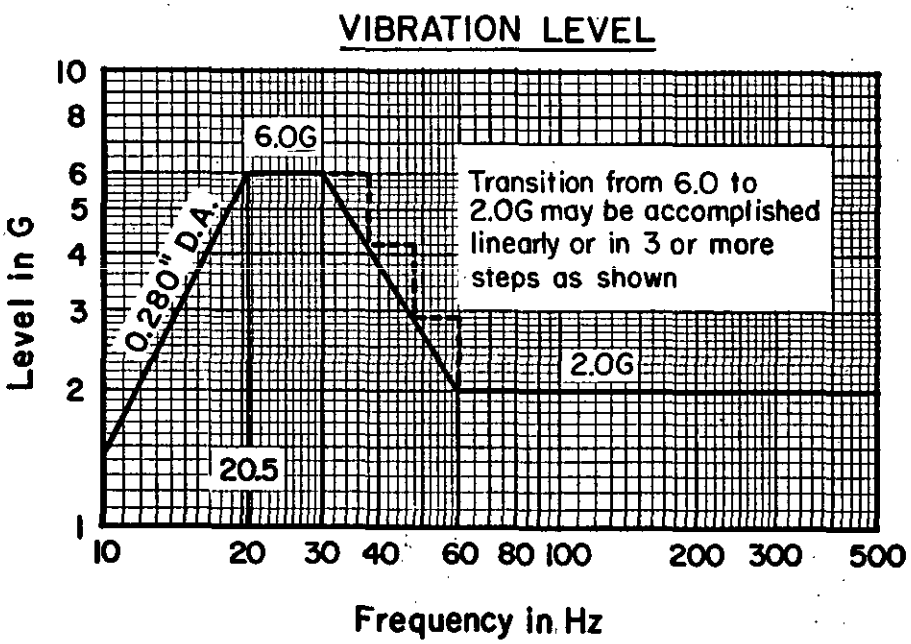
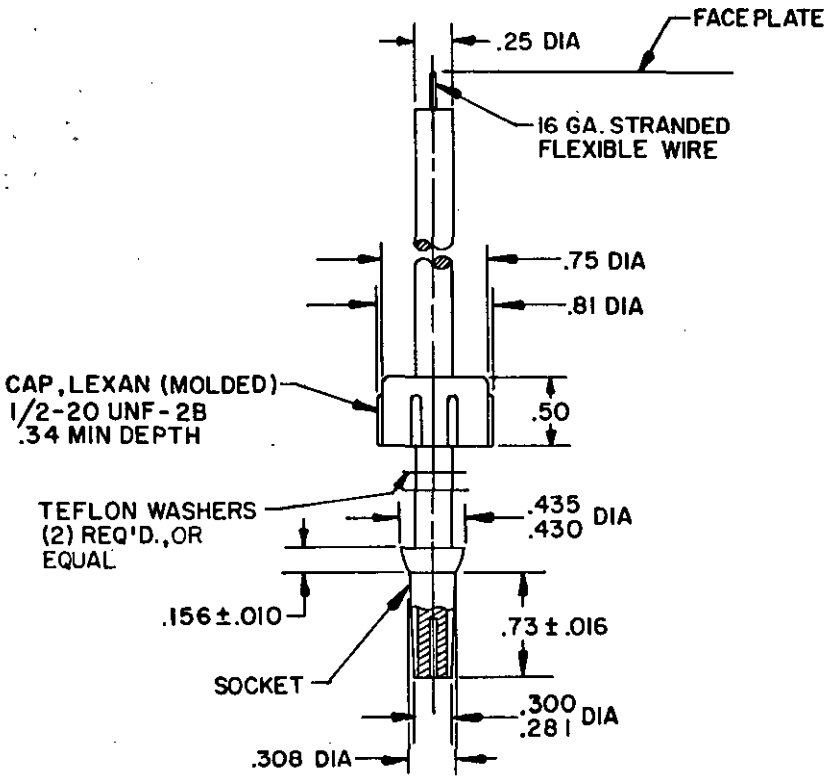


FIGURE 4. Vibration requirements.

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INCHES	MM	INCHES	MM
.010	.25	.430	10.92
.016	.41	.435	11.05
.156	3.96	.50	12.70
.25	6.35	.73	18.54
.281	7.14	.75	19.05
.300	7.62	.81	20.57
.308	7.82		
.34	8.64		

NOTES:
a. Dimensions are in inches.
b. Unless otherwise specified, tolerances are .005 (.13 mm) for three place decimals and .02 (.51 mm) for two place decimals.

FIGURE 5. Screen connection.

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